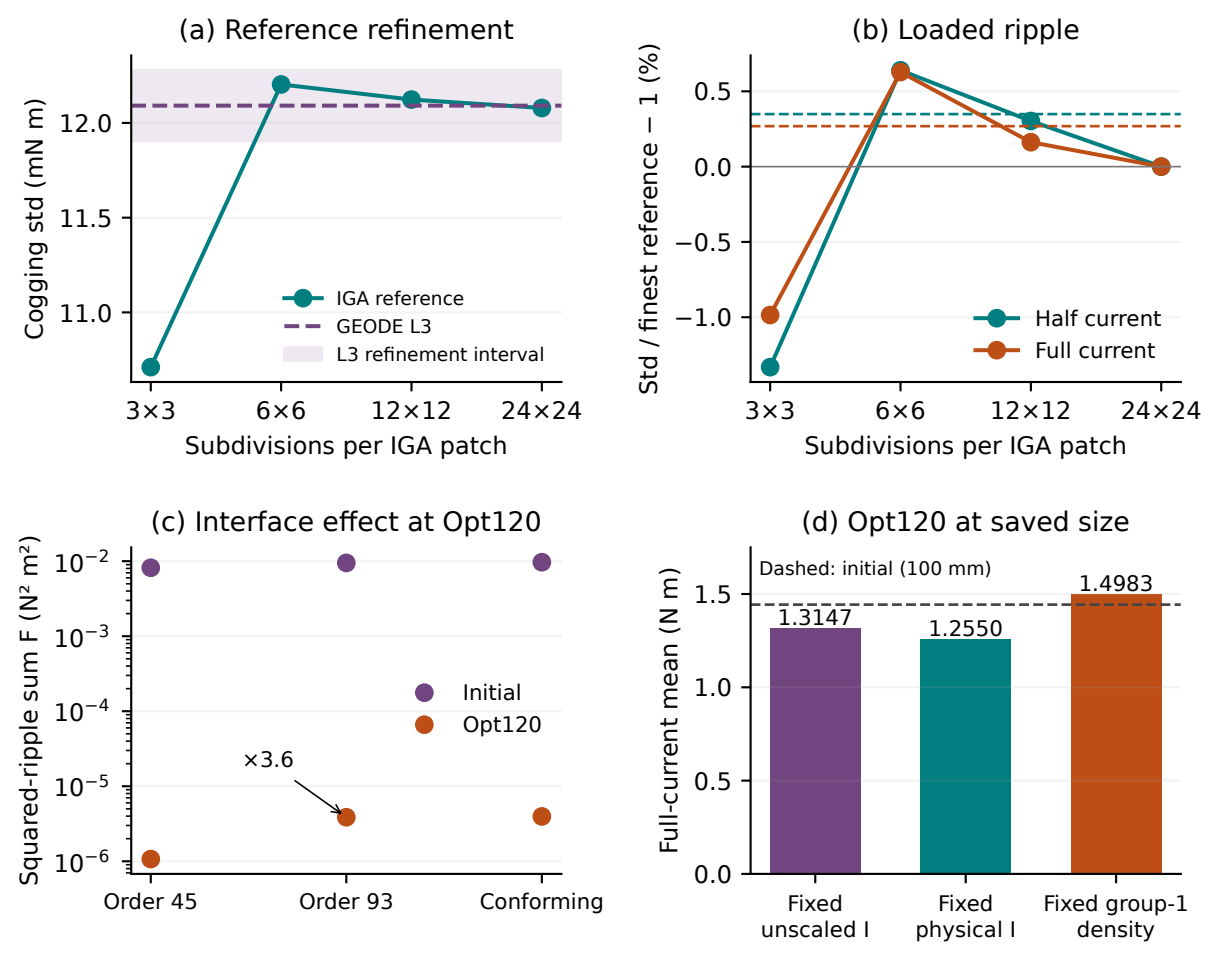
(a) Reference refinement
Cogging std (mN m)
12.0
11.5
11.0
IGA reference
GEODE L3
L3 refinement interval
3×3
6×6
12×12
24×24
Subdivisions per IGA patch
(b) Loaded ripple
Std / finest reference − 1 (%)
0.5
0.0
−0.5
−1.0
Half current
Full current
3×3
6×6
12×12
24×24
Subdivisions per IGA patch
(c) Interface effect at Opt120
Squared-ripple sum F (N² m²)
10⁻²
10⁻³
10⁻⁴
10⁻⁵
10⁻⁶
Initial
Opt120
×3.6
Order 45
Order 93
Conforming
(d) Opt120 at saved size
Full-current mean (N m)
Dashed: initial (100 mm)
1.5
1.0
0.5
0.0
1.3147
1.2550
1.4983
Fixed unscaled I
Fixed physical I
Fixed group-1 density